25/02/2025

Problem Set 2

Part (1) (10 points)

At MT1 the underlined problems must be done and turned in for grading. The others are some suggested choices for more practice.

A listing like §1B: 2, 5b, 10" do the indicated problems from supplementary problems section B.

1) Matrices and inverse matrices

- §1F: 5b, 8a; others: 5a, 9
- §1G: 3, 4, 5j

2) Theorems about square system: Equation of Planes

- §1H: 3a, 6c; others: 7
- §1E: 1c, d, 2, b; other 1a, b, e, 3, 5

Part II (17 points)

Problem I (6.1, 2 (2))

Suppose we know that when the three planes P_1, P_2, and P_3 intersect in pairs, we get three lines L_1, L_2, and L_3 which are distinct and parallel.

(a) Sketch a picture of the situation.

(b) Show that the three normals to P_1, P_2, and P_3 all lie in one plane using a geometric argument.

(c) Show that the three normals to P_1, P_2, and P_3 all lie in one plane using an algebraic argument.

(Why 3 planes do not intersect.)

Problem II (6.3: 1, 2)

A manufacturing process mixes three raw materials M_1, M_2, M_3 to produce three products P_1, P_2, and P_3. The ratios of the amounts of the raw materials (in the order M_1, M_2, M_3) which are used to make up each of the three products are as follows:

For P_1 the ratio is $1 : 2 : 3$.

For P_2 the ratio is $1 : 3 : 5$, and for P_3 the ratio is $3 : 5 : 8$.

In a certain production run, 137 units of M_1 and 279 units of each of the products P_1, P_2, P_3 were produced in that run.

(a) Set this problem up in matrix form. Use the letter A for the matrix and write down the (one-line) formula for the solution in matrix form.

(b) Compute the inverse of A and use it to solve for the production vector $\mathbf{P}$.

(c) Find a choice for the ratios for the third product (in lowest form), different from the other two ratios, for which the resulting system has non-unique solutions.

Problem 3 (6.4, 2)

For any plane P which is not parallel to the xy-plane, define the steepest direction on P to be the direction of any vector on P which makes the largest (acute) angle with the xy-plane.

Problem Set 2, Part II

1. (a) Let P be the plane through the origin with normal vector $\mathbf{n}$. Derive a formula, in terms of $\mathbf{n}$, for a vector $\mathbf{w}$ which points in the steepest direction on P.

   (b) Now let P be the plane through the origin which contains two non-parallel vectors $\mathbf{u}$ and $\mathbf{v}$, where $\mathbf{u}$ and $\mathbf{v}$ do not both lie in the xy-plane. Derive a formula, in terms of $\mathbf{u}$ and $\mathbf{v}$, for a vector $\mathbf{w}$ which points in the steepest direction on P.

Since $\mathbf{n}_1 \times \mathbf{n}_2$ is parallel to $\mathbf{n}_2 \times \mathbf{n}_3$, we can write one as a scalar multiple of the other.r.

$$\mathbf{x} \cdot \mathbf{n}_i = a_i$$

$$\begin{pmatrix} \mathbf{n}_1 \\ \mathbf{n}_2 \\ \mathbf{n}_3 \end{pmatrix} \mathbf{v} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

So the determinant of the matrix on the left must be zero, hence

$$0 = \det \begin{pmatrix} \mathbf{n}_1 \\ \mathbf{n}_2 \\ \mathbf{n}_3 \end{pmatrix} = \mathbf{n}_1 \cdot (\mathbf{n}_2 \times \mathbf{n}_3)$$

Therefore, the three normals are coplanar.

Q2(a)

In one unit of P_1, there are $\frac{1}{6}$ of a unit of M_1, $\frac{2}{6}$ of a unit of M_2, and $\frac{3}{6}$ of a unit of M_3.

So, to produce P_1, we need

$$\frac{1}{6}P_1 \text{ units of } M_1, \quad \frac{2}{6}P_1 \text{ units of } M_2, \quad \frac{3}{6}P_1 \text{ units of } M_3.$$

Using similar reasoning for P_2 and P_3, the linear system is

$$A\mathbf{p} = \mathbf{m}.$$

$$A = \begin{pmatrix} \frac{1}{6} & \frac{1}{9} & \frac{3}{16} \\ \frac{1}{3} & \frac{1}{3} & \frac{5}{16} \\ \frac{1}{2} & \frac{5}{9} & \frac{1}{2} \end{pmatrix}$$

(b)

$$|A| = \frac{1}{864} \neq 0 \quad \text{and hence } A^{-1} \text{ exists.}$$

$$\mathbf{p} = A^{-1}\mathbf{m}$$

$$= \begin{pmatrix} -6 & 42 & -24 \\ -9 & -9 & 9 \\ 16 & -32 & 16 \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}$$

If

$$\mathbf{m} = \langle 137,\ 279,\ 448 \rangle^T,$$

then

$$\mathbf{p} = A^{-1}\mathbf{m} = \langle 144,\ 288,\ 432 \rangle^T.$$

So, 144 units of P_1, 288 units of P_2, and 432 units of P_3 are produced.

(c)

$$A = \begin{pmatrix} \frac{1}{6} & \frac{1}{9} & \frac{1}{2}\left(\frac{1}{6} + \frac{1}{9}\right) \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{5}{9} & \frac{1}{2}\left(\frac{1}{2} + \frac{5}{9}\right) \end{pmatrix}$$